

一 解决问题一的 SAS 一般线性方程拟合程序

```
data DEMreg;
input net_h V;
cards;
3.41 0.834
3.42 0.835
3.49 0.837
3.5 0.839
3.57 0.84
3.64 0.843
3.68 0.844
3.79 0.849
4.01 0.857
4.13 0.861
4.51 0.876
7.51 1
8.06 1.02
9.48 1.12
10.25 1.139
23 1.8
25.81 2.054
27.33 2.055
27.56 2.071
27.91 2.094
27.97 2.12
28.23 2.116
28.85 2.159
29.08 2.175
29.23 2.185
29.3 2.19
29.75 2.222
29.97 2.237
30.51 2.328
31.55 2.351
31.66 2.389
31.82 2.4
32.58 2.457
33.13 2.469
;
proc glm data = demreg;
model v=net_h net_h*net_h*net_h;
output out=demregNew p=yhat r=resid stdr=eresid;
run;
proc gplot data = demregNew;
```

```

plot v*net_h = 1 yhat*net_h = 2 /overlay;
symbol1 c=blue i = none v=star;
symbol2 c=red i = spline v = none;
run;

proc glm data = demreg;
model v=net_h net_h*net_h net_h*net_h*net_h;
output out=demregNew p=yhat r=resid stdr=eresid;
run;

proc glm data = demreg;
model v= net_h*net_h net_h*net_h*net_h;
output out=demregNew p=yhat r=resid stdr=eresid;
run;

```

二、解决通过降水预测水位高程的 MATLAB 程序

```

%main
clear;
clc;
%data = load('C:\Documents and Settings\SCUT\桌面\data.txt');
a=[
2008-5-25    723  1.269    8   18
2008-5-26    725.3    1.3  10  25
2008-5-27    727.245  1.4823  0   0
2008-5-28    729.19   1.586   1   2
2008-5-29    730.13   1.61 2   5
2008-5-30    732.5    1.7725  0   0
2008-5-31    733.42   1.83 0   0
2008-6-1    734.68   1.9032  0   0
2008-6-2    735.86   2.01 0   1
2008-6-3    736.86   2.0864  0   0
2008-6-4    737.78   2.116   0   0
2008-6-5    738.93   2.19 12  12
2008-6-6    739.96   2.237   10  20
2008-6-7    740.67   2.372   0   2
2008-6-8    742.11   2.42 0   0
2008-6-9    742.58   2.457   0   0
2008-6-10    742.961  2.468525 0   0
2008-6-11    743.94   2.543525 0   0
2008-6-12    744.894  2.618525 0   0

];
%forcast= rQ(data(:,1));%计算预测数据

```

```

data = a(:,2:5);
high= data(:,1);
Q = data(:,2);
minRain=data(:,3)*0.001;
maxRain=data(:,4)*0.001;
[m,n]=size(data);
S=3550*10^6;%平方米
midRain=(minRain+maxRain)/2;
%midRain=minRain;
%16 hang
realHigh=high-710;
forcastH=zeros(m,4);
forcastH(1,:)=high(1);
tempQ=zeros(m,4);%当天的蓄量=昨天的蓄量+降水 4
deltaH=zeros(m,4);%当天的实际高度=反解方程组
tempQ(1,:)=Q(1);
sumof4=0;
alpha=0.9;
for i=2:m
    %tempQ(i)=Q(i)+midRain(i-1);
    if i<4
        sumof4=sum(midRain(1:i,1))/5;
    else
        sumof4=sum(midRain(i-3:i))/5;
    end
    if i<17
        tempQ(i,1)=tempQ(i-1,1)+alpha*sumof4*S/100000000*0.5;
        tempQ(i,2)=tempQ(i-1,2)+alpha*sumof4*S/100000000*0.8;
        tempQ(i,3)=tempQ(i-1,3)+alpha*sumof4*S/100000000*1;
        tempQ(i,4)=tempQ(i-1,4)+alpha*sumof4*S/100000000*1.5;
    else
        tempQ(i,1)=tempQ(i-1,1)+0.075;
        tempQ(i,2)=tempQ(i-1,2)+0.075;
        tempQ(i,3)=tempQ(i-1,3)+0.075;
        tempQ(i,4)=tempQ(i-1,4)+0.075;
    end
end
end

deltaH(1,:)=realHigh(1);
for i=2:m
    for j=1:4
        deltaH(i,j)=newton(Q(i),tempQ(i,j));
        forcastH(i,j)=710+deltaH(i,j);
    end
end

```

end

hold on

```
subplot(2,2,1),plot(1:m,forecastH(:,1),'b-',1:m,high,'r*',0:m,752*ones(m+1,1),'k-.'),legend('forecast','source'),title('50%'),ylabel('H(t)')
subplot(2,2,2),plot(1:m,forecastH(:,2),'b-',1:m,high,'r*',0:m,752*ones(m+1,1),'k-.'),legend('forecast','source'),title('80%'),ylabel('H(t)')
subplot(2,2,3),plot(1:m,forecastH(:,3),'b-',1:m,high,'r*',0:m,752*ones(m+1,1),'k-.'),legend('forecast','source'),title('100%'),ylabel('H(t)')
subplot(2,2,4),plot(1:m,forecastH(:,4),'b-',1:m,high,'r*',0:m,752*ones(m+1,1),'k-.'),legend('forecast','source'),title('150%'),ylabel('H(t)')
%plot(0:m,752*ones(m+1,1),'k-.')
%normrnd(1,0.5,100,1)
```

三、溃决模型的 MATLAB 程序

```
%main2
clear
clc
Q=[];
Q(1)=200;
%Q(1)=overtop;%漫顶流量
T=240;
deltaT=10/60;
Ntime=T/deltaT;
b=[];
b(1)=expand(0,48);
i=2;
H=[];
H(1)=12;%缺口深
V=[];
V(1)=2.457;%容量
while (i<Ntime)
    b(i)=expand((i-1)*deltaT,48);
    if Q(i-1)>0
        V(i)=0.09*deltaT/24+V(i-1)-Q(i-1)*deltaT*3600/10^8;
    else
        V(i)=V(i-1)+0.09*deltaT/24;
    end
    if (V(i)<0)
        break;
    end
    H(i)=erosion(H(i-1),V(i-1),Q(i-1),b(i));
    Q(i)=flood(H(i),b(i),V(i),V(i-1));
    if H(i)-H(i-1)<1e-9
```

```

        break;
    end

    i=i+1;
end
mm=floor(min(size(Q,2),210)/6)*6;
hold on
subplot(2,2,1),plot([6:mm]/6,Q(6:mm)),title('溃口流量 Q')
subplot(2,2,2),plot([6:mm]/6,H(6:mm)),title('溃口深度 H')
subplot(2,2,3),plot([6:mm]/6,V(6:mm)),title('水库蓄水量 Hw')
subplot(2,2,4),plot([6:mm]/6,b(6:mm)),title('溃口展宽 B')

```

四、模拟洪水演进的 MATLAB 程序

```

function [y,qq]=main3(Q);
%已经存在 Q 的行向量
m=600*length(Q);%10min
q=[];%zeros(1,m);
temp=[];

for j = 1:length(Q)
    temp=Q(j)*ones(1,600);
    q=[q temp];
end
%q;%生成流量 by second
qlt=zeros(1,200000);
h=zeros(1,200000);
dl=5;%m
v=5;
B=100;
dt=dl/v;
l=0;
t=0;
qm=max(q);
i=1;
while(i<200000)
    if i<=m
        l=l+dl;
        v=vl(l);
        dt=dl/v;
        t=t+dt;

        qlt(i)=flowdown(l,t,dt,q(i),qm);
    end
    i=i+1;
end

```

```

        h(i)=qlt(i)/v/B;

        i=i+1;
    else
        l=l+dl;
        v=vl(l);
        dt=dl/v;
        t=t+dt;
        if (abs(h(i)-h(i-1))<0.1)
            break;
        end
        qlt(i)=0.9*qlt(i-1);
        h(i)=qlt(i)/v/B;
        i=i+1;
    end
end
end
y=h;
qq=qlt;

```

五、溃决模型的 MATLAB 程序之二

```

function [y,qq,L,tt]=main33(Q);
%已经存在 Q 的行向量
m=120*length(Q);%10min
q=[];%zeros(1,m);
temp=[];
tt=zeros(1,200000);
L=zeros(1,200000);

for j = 1:length(Q)
    temp=Q(j)*ones(1,120);
    q=[q temp];
end
%q;%生成流量 by second
qlt=zeros(1,200000);
h=zeros(1,200000);
%dl=5;%m
v=5;
B=100;
dt=600;%
dl=v*dt;
l=0;

```

```

t=0;
qm=max(q);
i=1;
Wt=0;
while(i<200000)
    if i<=m
        l=l+dl;L(i)=l;
        v=vl(l);
        dt=dl/v;
        t=t+dt;tt(i)=t;
        Wt=Wt+q(i)*dt;
        %qlt(i)=flowdown(l,t,dt,q(i),qm);
        h(i)=Wt/l/B;
        qlt(i)=Wt/t;
        i=i+1;
    else
        l=l+dl;L(i)=l;
        v=vl(l);
        dt=dl/v;
        t=t+dt;tt(i)=t;
        if (abs(h(i)-h(i-1))<0.1)
            break;
        end
        h(i)=Wt/l/B;
        qlt(i)=Wt/t;
        i=i+1;
    end
end
end
y=h;
qq=qlt;

```

六、宽顶堰流量模型

```

function y=flood(h,b,v,v0)
%计算有缺口的流量
%h 当前深度
%b 当前溃口宽度
%v,v0 当前库容，上次库容
%bm=70;
bm=300;
hh=h-(42-newton(v0,v));
if hh>0
    y=1.71*b*hh^(1.5)+1.2*hh^(2.5)*h/((bm-b)/2);
    %y=1.71*b*hh^(1.5)+1.2*hh^(2.5)*h/((bm-b)/2);
    %4.5

```

```

else
    y=0;
end

```

七、溃口底部扩展模型

```

function b=expand(t,T)
%计算 t 时刻的溃口宽度
rou=1.5;
b0=40;
Bm=70;
%b=40+(Bm)*(t/T)^rou;
b=b0+(Bm-b0)*(t/T)^rou;
%b=150+(300-150)*(t/T)^rou;
%b=300*(t/T)^rou;

```

八、溃口侵蚀模型

```

function y=erosion(H,V,Q,B)
%H H(i)缺口深度
%V H(i)库容
%Q Q(i)流量（上一个）
%B B(i)上一个溃口底宽
H_water=newton(20,V);
H_water_gc=710+H_water;
deltaH=H_water_gc-752+H;
Pb=2*deltaH+B;
Lb=803;
deltaT=10/60;
p=0.1;%空隙率 1-p
ks=0.01;%运输效率 19
y=H+Q*deltaT*3600*ks/(Pb*Lb*(1-p));

```

九、自然漫顶模型

```

function Q=overtop()
%返回初始的流量
Nr=760;%漫过高程
Nc=753;%坝顶高度
B=300;
lc= 803;
g=9.8;
c=1/3+1.32*(Nr-Nc)/B;

```

$Q = c * l_c * \sqrt{2 * g} * (N_r - N_c)^{(3/2)}$;

九、自然漫顶过程模拟的程序

```
%1/3
clear
clc
Q=[];
Q(1)=23000;
%Q(1)=overtop;%漫顶流量
T=240;
deltaT=10/60;
Ntime=T/deltaT;
b=[];
b(1)=expand(0,48);
i=2;
H=[];
H(1)=30;%缺口深
V=[];
V(1)=2.457;%容量
while (i<Ntime)
    b(i)=expand((i-1)*deltaT,48);
    if Q(i-1)>0
        V(i)=0.09*deltaT/24+V(i-1)-Q(i-1)*deltaT*3600/10^8;
    else
        V(i)=V(i-1)+0.09*deltaT/24;
    end
    if (V(i)<0)
        break;
    end
    H(i)=erosion(H(i-1),V(i-1),Q(i-1),b(i));
    Q(i)=flood(H(i),b(i),V(i),V(i-1));
    if H(i)-H(i-1)<1e-9
        break;
    end

    i=i+1;
end
mm=floor(min(size(Q,2),210)/6)*6;
hold on
subplot(2,2,1),plot([6:mm]/6,Q(6:mm)),title('溃口流量 Q')
subplot(2,2,2),plot([6:mm]/6,H(6:mm)),title('溃口深度 H')
subplot(2,2,3),plot([6:mm]/6,V(6:mm)),title('水库蓄水量 Hw')
subplot(2,2,4),plot([6:mm]/6,b(6:mm)),title('溃口展宽 B')
```

十、newton 法解方程

```
function [x1,n]=newton(x0,q)
```

```
x1=x0 - Q(x0,q)/dQ(x0);
```

```
n=1;
```

```
while(abs(x1-x0)>=1.0e-9)&&(n<=1000000)
```

```
    x0 = x1;
```

```
    x1 = x0 - Q(x0,q)/dQ(x0);
```

```
    n=n+1;
```

```
end
```